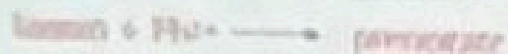


2.2 ISOLATION OF ACTIVE CONSTITUENTS

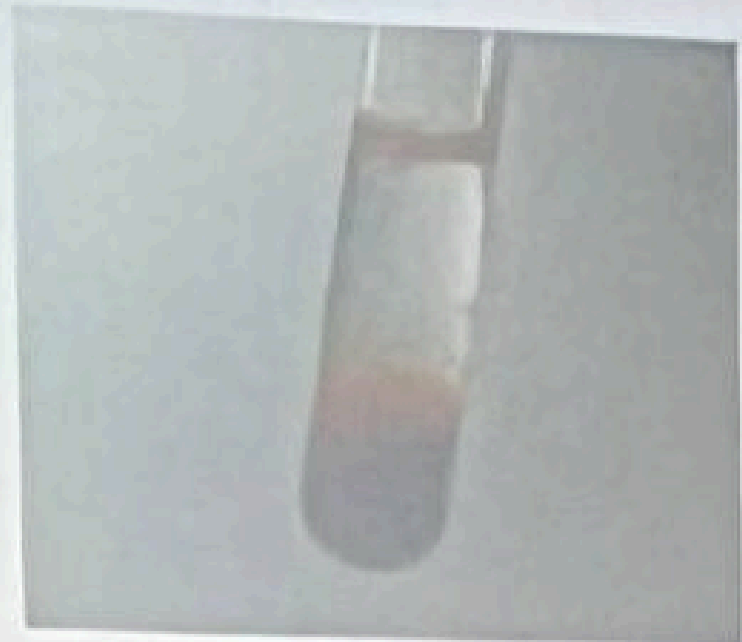
2.2.1 Classical Methods

A. Precipitation

- A **solid** formed by a change in a solution
 - Due to a chemical reaction



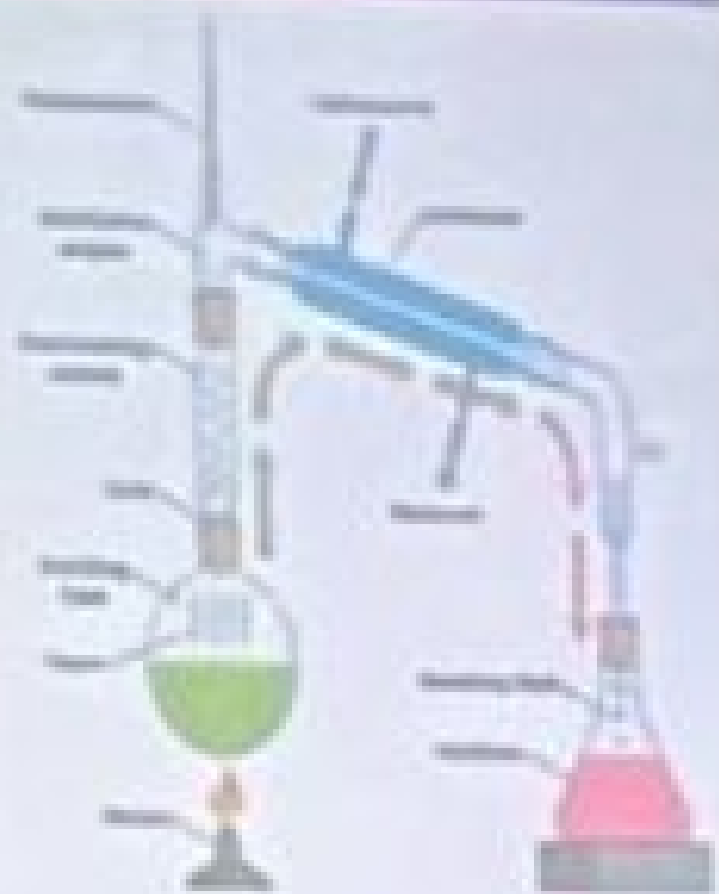
- Altering the Ph
 - Precipitation of proteins
- Change in temperature that decreases solubility of a solid
 - Pure compounds



B. Fractional distillation

- Separation of miscible liquids based on the **difference in boiling point** of compounds
- Eg. Different solvent mixture can be separated

FRACTIONAL DISTILLATION



C. Formation of chemical derivatives

- Separation by reversible reactions which can alter certain property of a substance
 - Phenolic compounds which are organic soluble are treated with alkalis they form water-soluble phenolate salts. Phenolate salts can generate back the original phenolic compounds if treated with acids.
 - Carboxylic acids can be made volatile by esterification and reversed by hydrolysis.

D. Crystallization

- Crystallization differs from precipitation in that the product is a well-defined crystalline substance
- Crystallization occurs in supersaturated solutions and Achieved by
 - Rapid cooling
 - evaporation of solvent
 - salting out
 - adding a solvent in which the compound of interest is not soluble in

E. Sublimation

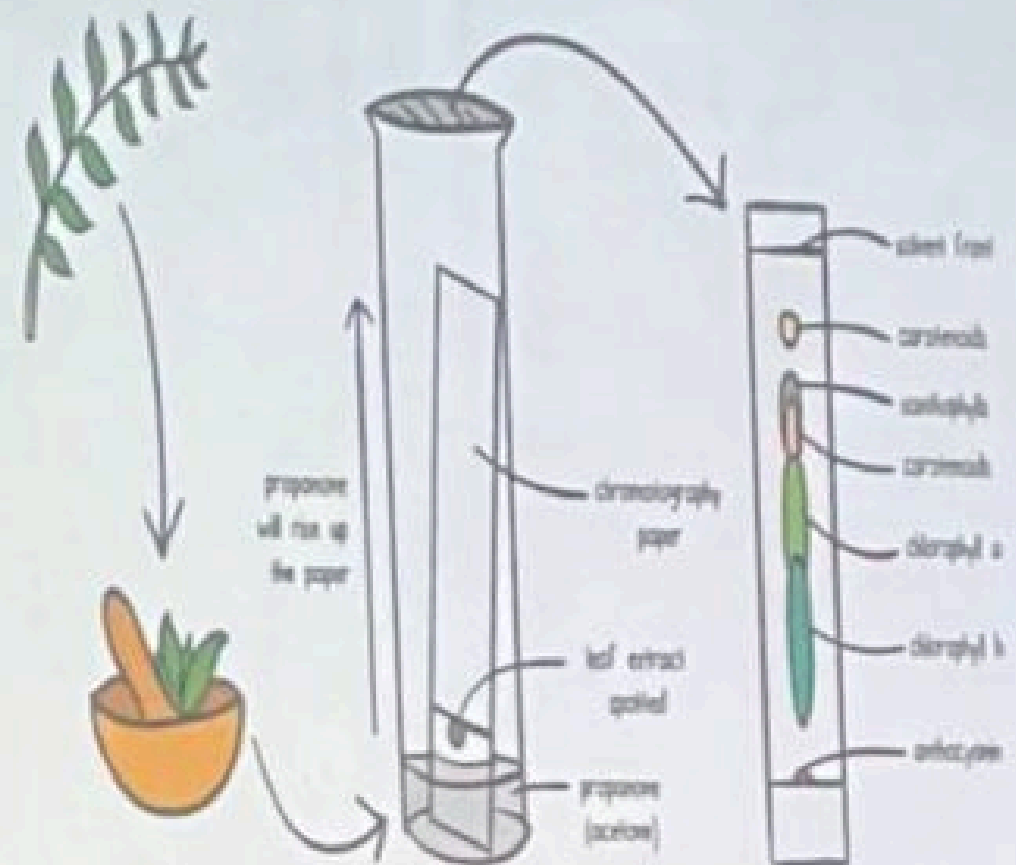
- Sublimation is when a substance in the **solid phase** passes directly into the **vapor phase** without going into the liquid phase
 - Eg. **Caffeine** from dried tea leaves

2.2.2 Modern separation techniques

Chromatography

IUPAC defines chromatography as:

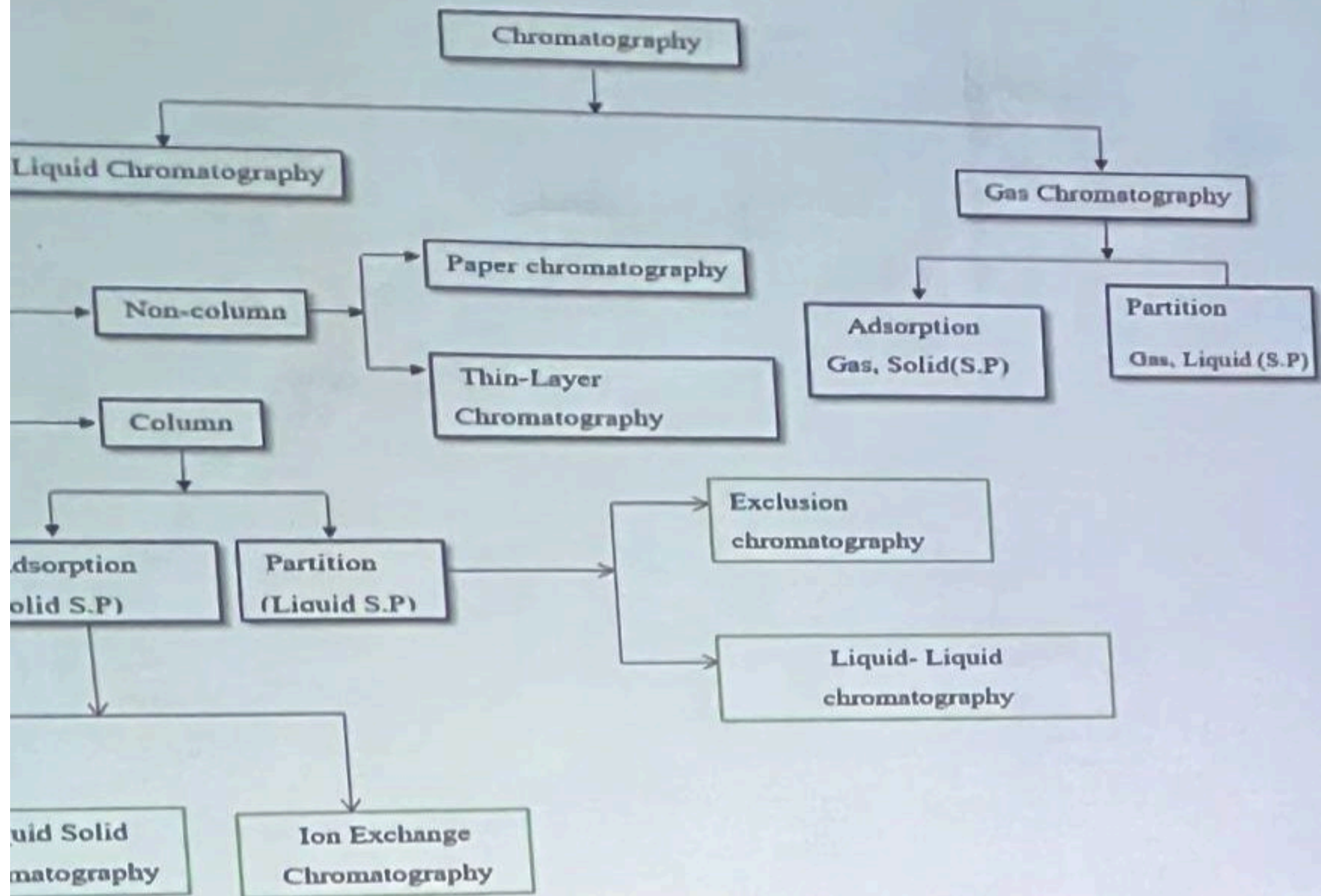
"Chromatography is a physical method of separation in which the components to be separated are distributed between two phases, one of which is **stationary** (stationary phase) while the other (the **mobile phase**) moves in a definite direction."



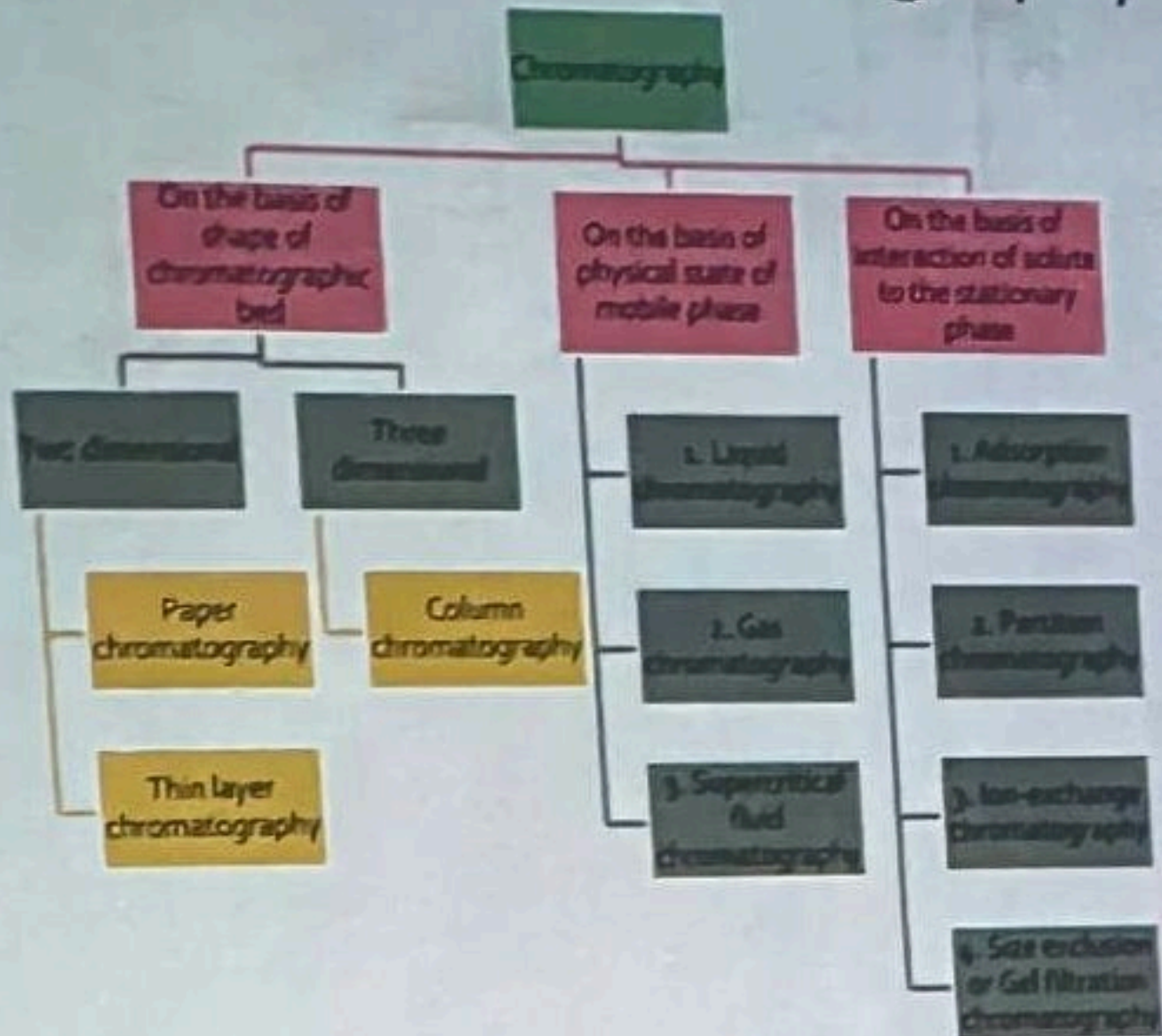
Principles of Chromatography

- The mobile phase may be a gas or a liquid (solvent system) and the stationary phase may be a liquid film on the surface of an inert support material or a solid surface.
- Interaction occurs between the solute and stationary phase so that the solute is distributed between the stationary phase and mobile phase.

- Attraction of the solute for the stationary phase results in retardation of its movement through the chromatography system.
- Different components (solutes, compounds) will move at differing rates since each will have a slightly different affinity for the stationary phase with respect to the mobile phase.
- The stationary phase may be a solid or a liquid on the surface of an inert support material and the fluid mobile phase (solvent system) may be liquid or gas, to give liquid chromatography and gas chromatography, respectively.



Classification of Chromatography



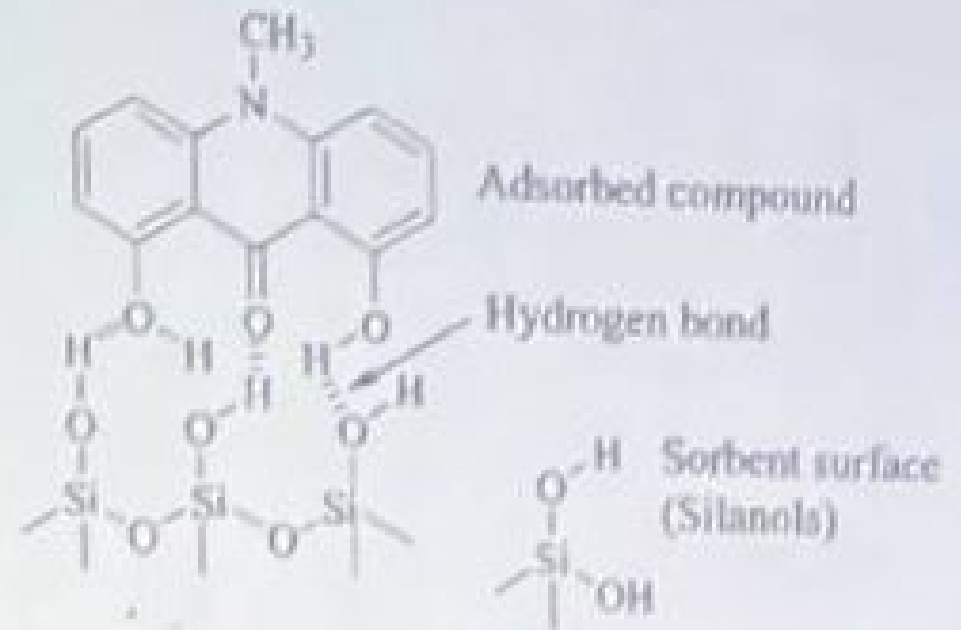
Classification According to the Mechanism of Separation

Based on Mechanism of separation

Adsorption Chromatography

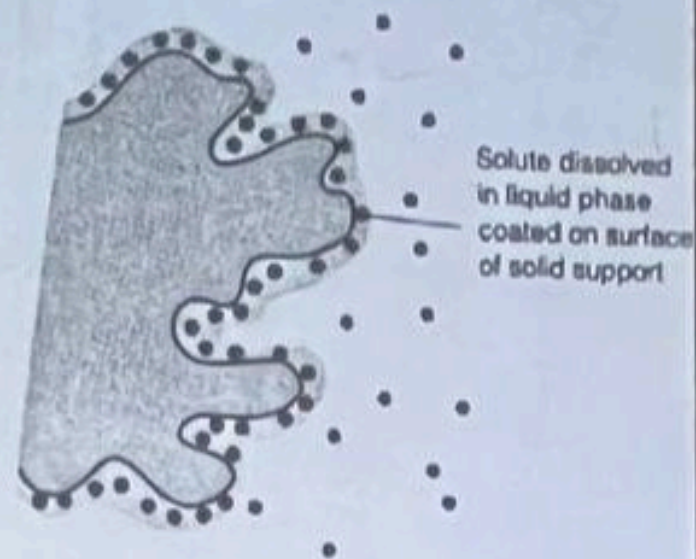
- Oldest types of chromatography around.
- The equilibration between the mobile (liquid or gas) and stationary phase (solid) accounts for the separation of different solutes.
- Components of a mixture with better affinity to the stationary phase (polarity solutes) will be adsorbed strongly; hence, their movement will be impaired greatly.
- Components of the mixture with lesser affinity to the stationary phase (less polarity solutes) will move with the mobile phase at relatively faster rates.

- A solvent system is composed of both **polar** (major portion and used to dissolve and carry away solute molecules) and **non-polar** portion (minor portion and used to displace solute molecules)
- It is always normal phase chromatography



Partition Chromatography

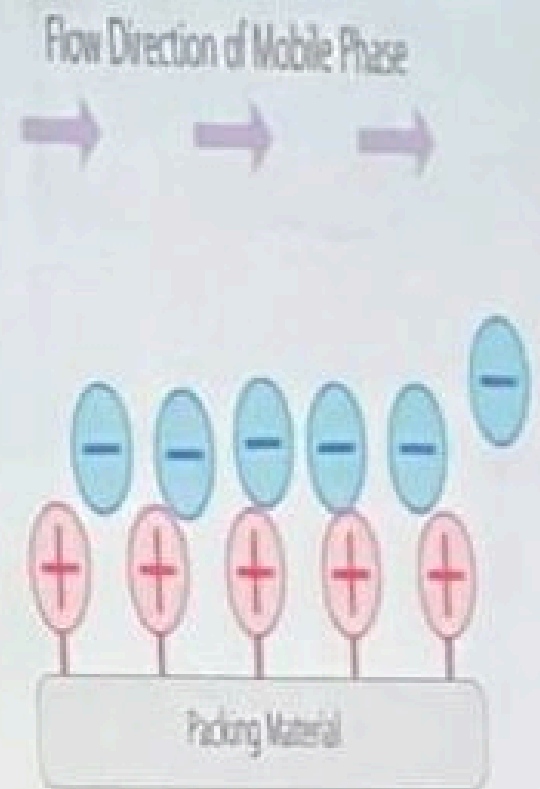
- This form of chromatography is based on a **thin film formed on the surface of a solid support** (usually silica gel and cellulose) by a liquid stationary phase.
 - Eg. Long chain hydrocarbons attached to a solid silica gel
- Solute equilibrates between the mobile phase and the stationary liquid and **separation is due to difference in the partition coefficients or relative solubility** of the various components of mixture.
- In most cases mixtures of compounds with known or predictable solubility are separated by this method. E.g. compounds belonging to a specific chemical class.



Partition chromatography

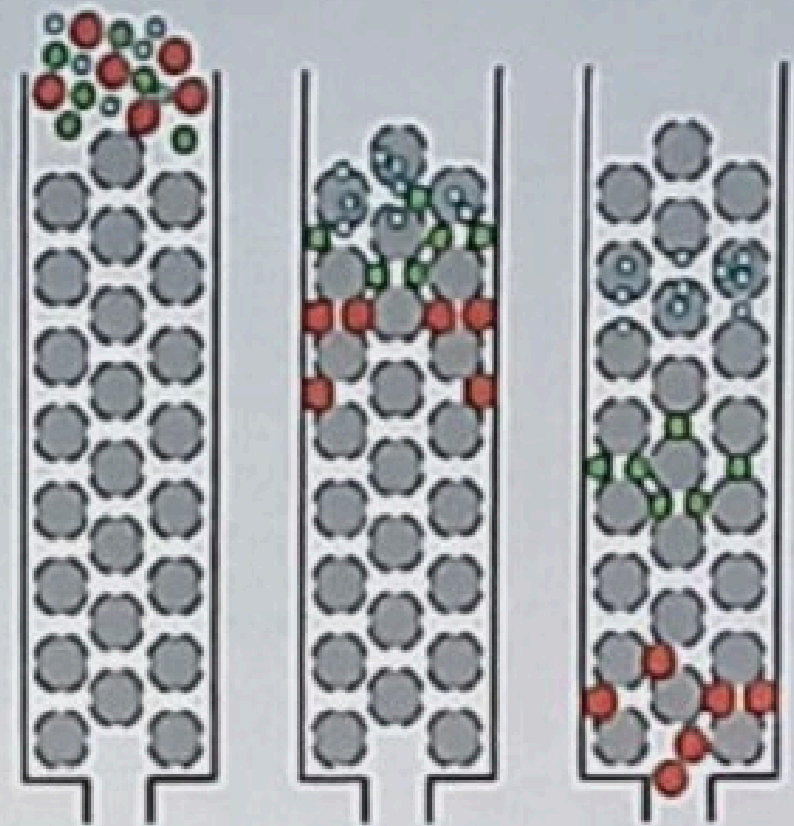
Ion Exchange Chromatography

- use of a resin (the stationary solid phase) is used to covalently attach anions or cation onto it.
- Solute ions of the opposite charge in the mobile liquid phase are attracted to the resin by electrostatic forces.
- **Cation exchangers** commonly have acidic functional groups such as: SO_3H , $-\text{PO}_4\text{H}_2$, $-\text{CO}_2\text{H}$, etc.
- **Anion exchangers** commonly have tertiary or quaternary ammonium groups: e.g. $-\text{N}^+(\text{CH}_3)_2 \text{HCl}^-$, $-\text{N}^+(\text{CH}_3)_3 \text{Cl}^-$
- The mobile phase is always an aqueous solution which is usually buffered to a particular pH. It also contains some amount of the exchangeable ion.
- **Common applications:** Water treatment, Separation of amino acids



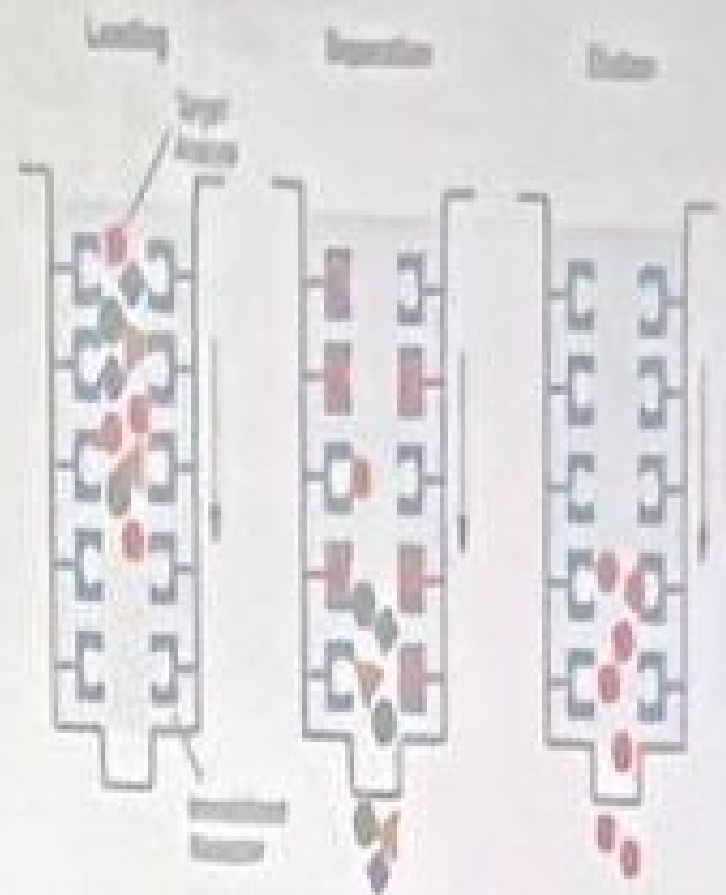
Molecular Exclusion Chromatography

- Also known as **gel permeation** or **gel filtration**, this type of chromatography lacks an attractive interaction between the stationary phase and solute.
- The liquid or gaseous phase passes through a **porous gel**, which **separates the molecules according to its size**.
- The pores are normally small and exclude the larger solute molecules, but allow smaller molecules to enter the gel, causing them to flow through a larger volume.
- This causes the **larger molecules** to pass through the column at a **faster rate** than the **smaller ones**.
- Recovery of solute is done via altering the PH of the gel matrix



Affinity Chromatography

- This is the most selective type of chromatography employed.
- It utilizes the specific interaction between one kind of **solute molecule** and a **second molecule** that is **immobilized** on a stationary phase.
- **Stationary phase** is a pocket type protein that binds **ligand temporarily**
- Eg. The immobilized molecule may be an antibody to some specific protein.
 - When solute containing a mixture of proteins is passed by this molecule, only the specific protein is reacted to this antibody, binding it to the stationary phase.
 - This protein is later **eluted** by changing the ionic strength or pH.



Based on Nature of the Mobile Phase

- Liquid chromatography
- Gas chromatography
- Supercritical chromatography

Based on the Polarity of the Stationary Phase

- **Normal phase chromatography:** the stationary phase is polar substance.
- **Reversed phase chromatography:** the stationary phase is non-polar substance.

Based on the purpose of the chromatographic process

- **Preparative chromatography** seeks to separate the components of a mixture for further use (and is thus a form of purification).
- **Analytical chromatography** normally operates with smaller amounts of material and seeks to measure the relative proportions of analytes in a mixture. The two are not mutually exclusive.

Based on the Configuration of Instrument

- **Planar chromatography**
TLC, PC...
- **Column chromatography**
GC, HPLC, FCC,

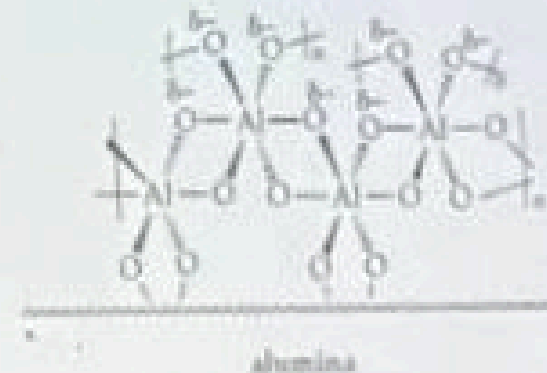
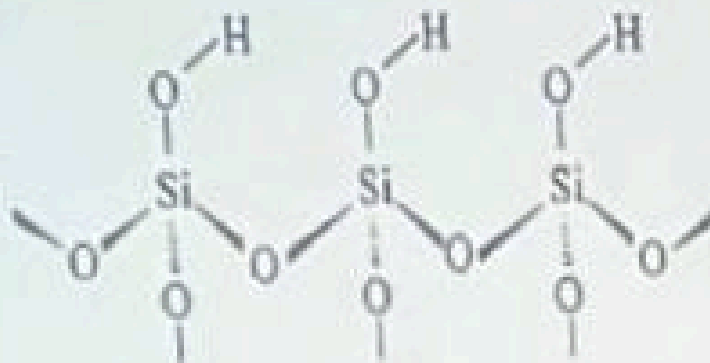
Planar Chromatography

- Planar chromatography is a separation technique in which the stationary phase is present as or on a plane.
- The plane can be:
 - A paper, serving as such or impregnated by a substance as the stationary bed (**paper chromatography**)
 - A layer of solid particles spread on a support such as a glass or aluminum plate (**thin layer chromatography**).

Thin Layer Chromatography (TLC)

Thin layer chromatography (TLC) is a form of planar chromatography that involves:

- A **stationary phase** of a thin layer of adsorbent material, **usually silica gel, aluminium oxide, or cellulose immobilised onto a flat, inert carrier sheet.**
- A **liquid phase** consisting of the solution to be separated is then dissolved in an appropriate solvent and is drawn up the plate **via capillary action,**
- **Separating is based on the polarity of the analyte.**



- Different compounds in the sample mixture travel different distances according to how strongly they interact with the adsorbent (**polarity**).
- TLC can be used
 - To separate components in a mixture (preparative TLC)
 - To assess purity (how many spots are present)
 - To identify a substance (analytical TLC)
 - To assess compound polarity

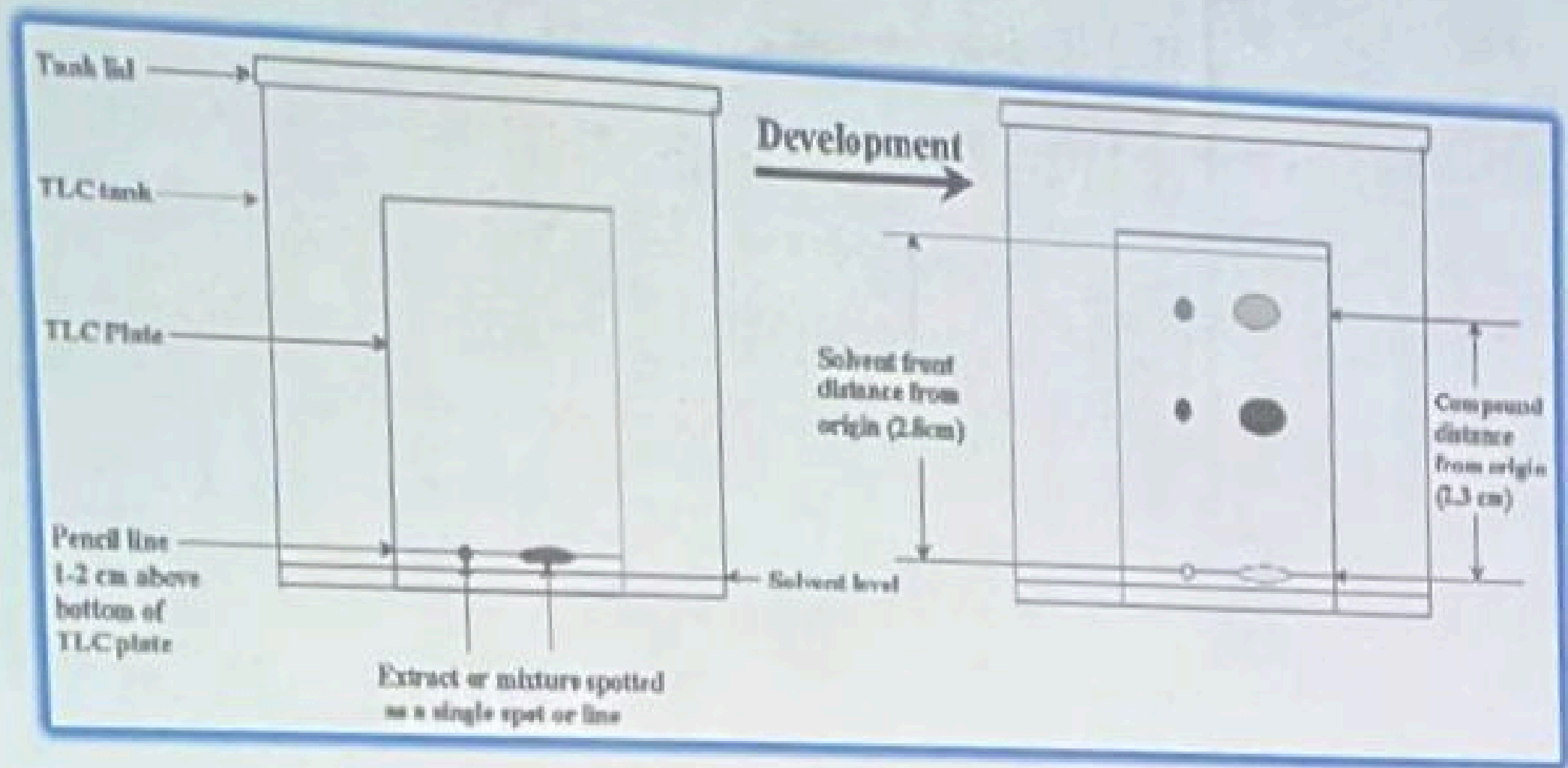
Steps in TLC

Stationary phase preparation

- TLC plates are made by mixing the *adsorbent*, such as *silica gel* or *Alumina* (polymer of *aluminum oxide* (Al_2O_3)), with a small amount of inert binder like *calcium sulfate* (gypsum) and water.
- A reversed phase is also available Eg. C10, C18
- This mixture is spread as thick slurry on an unreactive *glass*, *thick aluminum foil*, or *plastic*, and *activated* by drying in an oven for thirty minutes at 110°C
- The thickness of the adsorbent layer is typically around 0.1–0.25 mm for analytical purposes and around 1–2 mm for preparative TLC.
- Precoated plates are available on the market

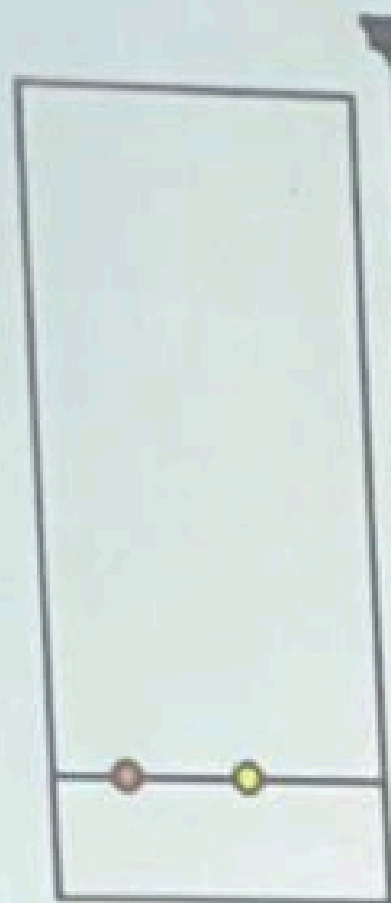
Sample application

- Prior to TLC, samples are dissolved in a suitable solvent:
 - It should be as volatile, less viscous and weak as possible.
- A small spot of solution containing the sample is applied to a plate, about 1.5 cm from the base on 20 x 20 cm plate.
- Application could be in form of spots (for analytical chromatography) or in form of bands (for preparative work) by using capillary tube, a micropipette or a calibrated glass microsyringe.
- Spots should not be closer than 1 cm center to center, should be 2-5 mm in diameter and 1-5 $\mu\text{g}/\mu\text{l}$ solution concentrations are practicable.



Development

- Separation by TLC is effected by the application of a mixture or extract as a spot or band on to a sorbent (TLC plate).
- The plate is then placed into a tank with sufficient suitable solvent **just to wet the lower edge of the plate/sorbent**
- Before development, the atmosphere in the tank is allowed **to saturate** with the **solvent system (mobile phase)** usually for 30 minutes to 1 hour.
 - To avoid skewed solvent front or development
- The solvent front then migrates up the plate through the sorbent **by capillary action**, and this process is **known as development**.



Spotted
TLC Plate

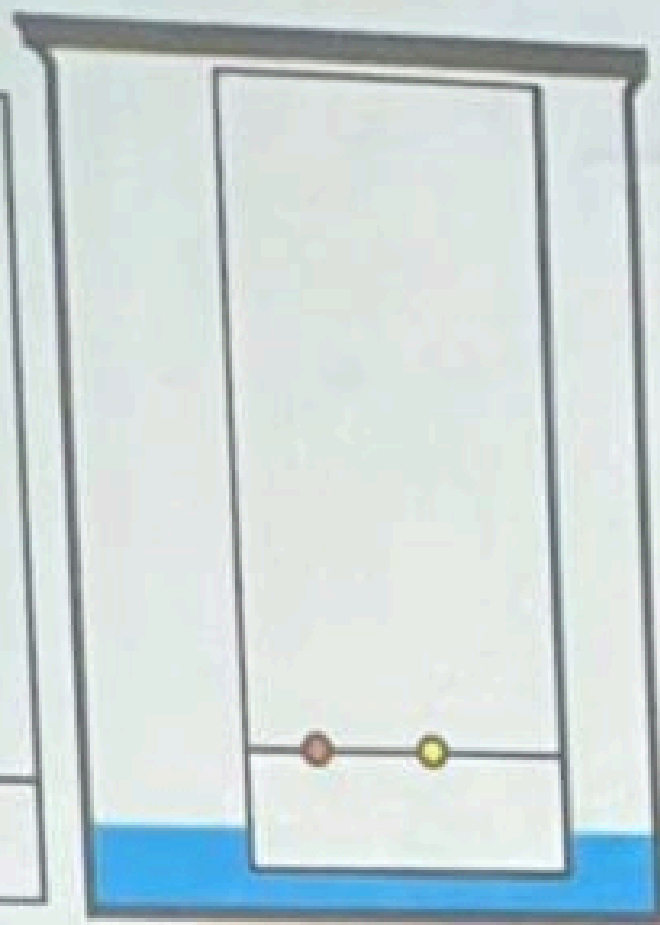
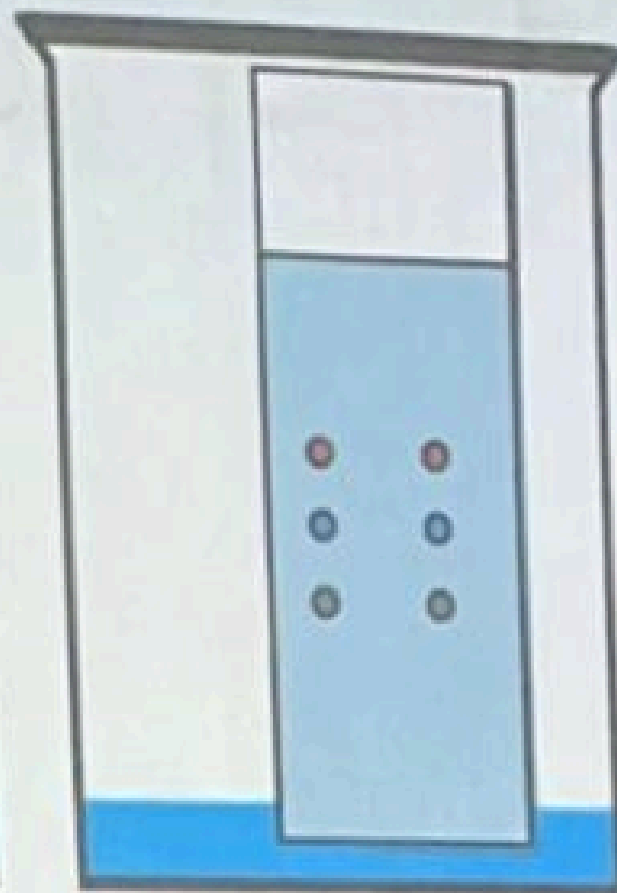


Plate inserted into
solvent saturated chamber



Developing Chamber

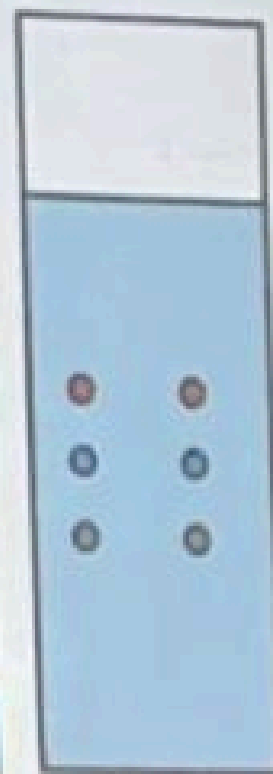
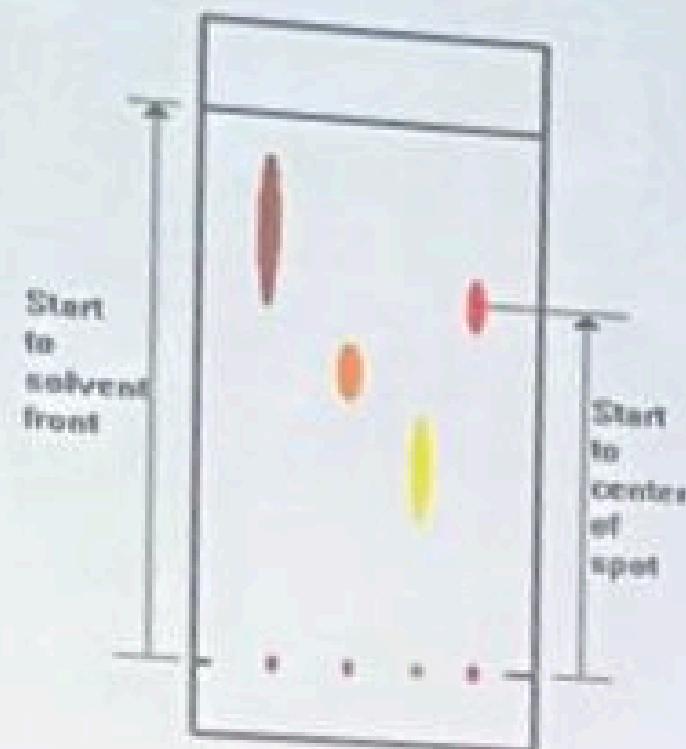


Plate with separated
component

- A factor in quantifying migration of a compound on a particular sorbent and solvent system is the **retention factor (R_f)** value which refers to migration 'relative to the solvent front'. This is defined as:

$$R_f = \frac{\text{Compound distance from origin (midpoint)}}{\text{Solvent front distance from origin}}$$

- R_f value can aid identification.
- **Ideal range is 0.3-0.7**
- What if the R_f value is 1 or 0?

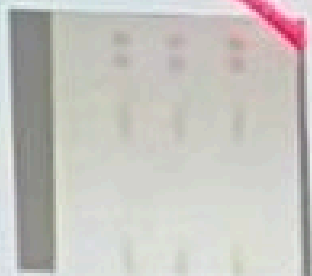


I. Analysis

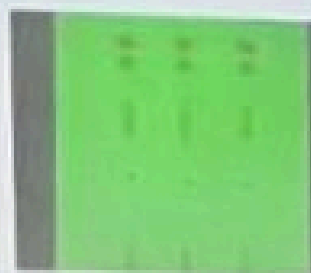
- As the chemicals being separated may be colorless, several methods exist to visualize the spots:

A. Physical detection methods

- Visible light
- UV light using a UV chamber



Commercial TLC plate after development in normal lighting.



Same TLC plate held under a UV lamp - Note the appearance of additional spots.

B. Chemical detection methods

Spraying reagents/Chromogenic reagents.

- **General reagents** e.g. H_2SO_4 for all organic compounds, I_2 for all unsaturated compounds etc.
- **Specific reagents** e.g. ~~Ni~~hydrin for aminoacids, Dragendorff's reagent for alkaloids, etc
 - Tests could result in **reversible or irreversible** reactions

Advantages and disadvantages of TLC

Advantages

- It is cost-effective
- Easy scale-up from analytical to preparative scale
- Flexibility of choice of mobile and stationary phases
- A separation may be readily optimized methods may be quickly developed
- A large number of samples may be analyzed or separated simultaneously.
- Does not require a regeneration step, since TLC and HPTLC plates are disposable items

Disadvantages

- It has a limited resolution power due to the short (3-10 cm) separation length.
- As an 'open' system, TLC depends on environmental factors
 - E.g. influence of relative humidity on hydrophilic layers (control necessary);
 - volatile, oxygen and light-sensitive samples.
- Lower reproducibility of retention values and other chromatographic parameters.
- Lengthy separation process.
- Greater difficulty and complexity of the process automatization.